

Reviewer Supplement: Artifact-Gated Evaluation of a Causal FIR Module for Sequential Recommendation

[MAINTAINER: REAL AUTHOR NAME(S) REQUIRED – TORS IS SINGLE-BLIND], [Maintainer: real institution required – TORS is single-blind], [Maintainer: country]

This reviewer supplement preserves the extended secondary-study record, negative results, protocol chronology, and appendix tables omitted from the length-focused main manuscript. It changes no verdict, claim boundary, or evidentiary status.

1 Extended supporting results

AlphaFuse pipeline boundary. Because the upstream offline preprocessing code was unavailable, both official classes consumed training rows produced by our frozen per-prefix adapter. This is upstream-class/training-code transfer, not reproduction of the upstream data pipeline.

Statistical reporting conventions. Unless stated otherwise: the sample unit is the training seed; means are reported $\pm$ sample standard deviation over seeds; confidence intervals are two-sided 95% Student-t intervals with $df = n-1$; comparisons between our own arms use same-numbered seeds and are analyzed as independent arms (Welch/Satterthwaite) unless exact matched backbone initialization and a paired analysis are explicitly named, as in canonical breadth, active controls, pointwise placebo, and prospective Software V3; comparisons against published numbers are point-estimate comparisons; pooled tail hit-counts are descriptive counts only; p-values are uncorrected unless a correction is named. Cells labeled *confirmatory* in the artifact manifest are pre-declared prospective campaigns or frozen-protocol locks: selection timing and protocol integrity, not seed count or external independence, are the criterion. Multi-seed post-hoc results remain exploratory.

1.1 The Musical_Instruments frequency-5 tail case (cross-dataset heterogeneity not established)

Beyond *whether* text helps overall (modestly: +2.7% on Video_Games, §5.1), we ask *where this additive frozen-text construction helps* (only early additive fusion of frozen text features is tested – §2.3 scope note). `evaluate()` reports by_popularity NDCG@10 over nominal train-frequency terciles {tail, mid, head} (bucketed by TRAIN frequency only, so no validation/test frequencies enter the bucketing; two cohort-definition defects are disclosed below), and we compare the full text stack against an ID-only ablation on the **same split and seed numbers**. The cold-start intuition – text should rescue rare items where ID embeddings are starved – holds on *one* of three datasets, and the MI difference survives an independent-arm analysis.

Randomization disclosure. Same-numbered text and ID arms are not initialization-paired: treatment-specific modules consume RNG before the shared backbone is constructed, changing backbone initialization and subsequent stochastic streams. Table 1d's per-seed differences are therefore descriptive, all retained inference is independent-arm Welch/Satterthwaite, and the former paired CI/MDE/TOST analyses are withdrawn.

Table 1d: text – ID tail-tercile NDCG@10 contrast (same-seed-number arms – NOT initialization-paired; per-dataset estimates – cross-dataset heterogeneity not established, point 3).

Author's Contact Information: [Maintainer: real author name(s) required – TORS is single-blind], [Maintainer: real institution required – TORS is single-blind], [Maintainer: country].

dataset	catalog density	tail Δ NDCG@10 (5-seed)	seeds pos- itive	verdict
Musical_Instruments	sparse	+0.000335 $\pm$ 0.000195 , Welch 95% CI [+0.000109, +0.000562] excl 0	5/5	frequency-5 tail advantage (independent-arm; §1.1)
Video_Games	dense	-0.000148 ± 0.000179	2/5	null — equivalence NOT established (retraction, point 2)
Beauty_and_PC	dense	-0.0000078 (3-seed, sample-sd 0.000020)	1/3	exploratory (3-seed; mixed eval geometry)

The historical MI contrast survives independent-arm analysis (Welch $t = 3.94$, $p = 0.014$, 95% CI [+0.000109, +0.000562]) and has tail HR $\Delta = +0.00109 \pm 0.00065$, but its absolute scale is small (≈ 29.6 versus 20.0 hits/seed); MI's largest absolute text lift is at the head (+0.00527). On Video_Games, no tail difference is detected (Welch $p \approx 0.22$), but equivalence is **not** established. Beauty remains exploratory because one arm pair uses unmatched evaluation geometry. Most importantly, the repaired TFFV2 MI-VG contrast is +0.000247 ($p = 0.13$), so cross-dataset heterogeneity did not replicate and is not claimed (Fig. 1).

Why Table 1d is historical. Its nominal terciles mix zero-train-exposure targets with positive-frequency targets and split boundary-frequency ties by item-ID order; the original runs also lack per-row outcome sidecars. These arm-independent defects do not create a treatment-specific cohort, but they prevent the intended rare-item interpretation. TFFV2 therefore separates zero exposure, uses whole-frequency-group cohorts, and releases per-user outcomes.

Repaired-estimand campaign (TFFV2; 8 independent seeds/arm). On the tie-safe MI positive-frequency tail, text – ID is **+0.000420**, 95% CI [**+0.000181**, **+0.000660**], Welch $p = 0.0022$, passing the frozen Holm rule. The [1,6] band (+0.000546, $p = 0.0039$) and HR@10 (+0.001150, $p = 4 \times 10^{-5}$) agree. However, zero-exposure targets produce no hits through rank 100 in any of four categories, the exclude-frequency-5 sensitivity is nonsignificant (+0.000071, $p = 0.52$), and the frequency-5 group alone is +0.001379. We therefore report a **frequency-5-heavy positive-tail case**, not cold-start capability or a smooth rare-item law. VG is nonsignificant (+0.000173, $p = 0.14$), as is MI-VG (+0.000247, $p = 0.13$).

Evidence status. TFFV2's endpoints were Git-frozen before launch, but interim test values became visible, its first independently verifiable timestamp postdates the first result, and the adjudicator was committed after some outputs existed. The eight-seed size was resource-based rather than prospectively powered, and the run manifests record concurrent tracked-document edits. TFFV2 is therefore **outcome-visible, not confirmatory**; its released sidecars and frozen estimator support reproducible estimates, not a claim of blinded confirmation. The repository retains the complete chronology, cohort files, logs, hashes, and errata.

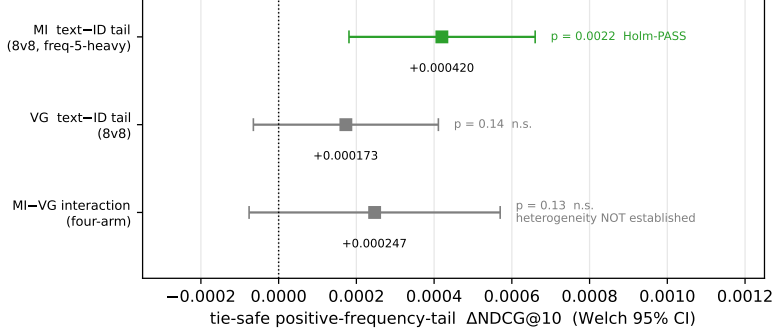
1.2 Titration: thinning tracks the head effect but not the tail effect (level contrasts under a bundled intervention; head trend exploratory and uncorrected; tail inconclusive)

We thinned Video_Games interactions over six rungs down to MI's global density ($\rho = 0.66$), holding n_{eval} at 94,762 and re-estimating text–ID differences with five seeds per rung. The two strata respond differently:

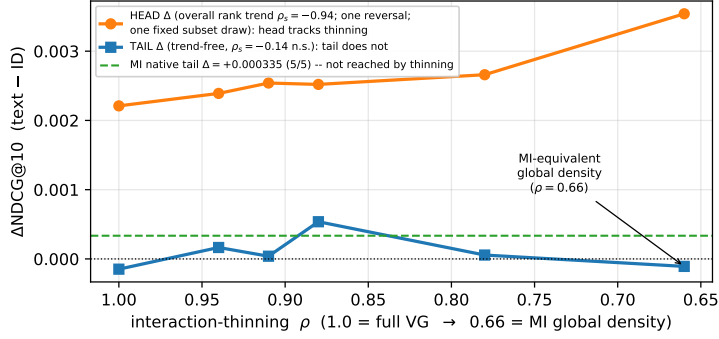
- **Head:** Δ rises overall from +0.00221 to +0.00354; all rungs are 5/5 positive. Spearman ρ_s versus density is -0.94 (exact $p = 0.017$), with one adjacent reversal; the HR trend is weaker ($\rho_s = -0.71$, $p \approx 0.14$). This is an exploratory uncorrected rank trend on one fixed thinning draw.

The MI frequency-5 tail case and its two-axis descriptive contrast
 (A: TFV2, 8 independent seeds/arm, outcome-visible; B-C: one fixed subset draw, 5-seed summaries, no draw uncertainty; heterogeneity not established, $p = 0.13$; AR2023 5-core LLOO, NDCG@10)

(A) TFV2 repaired-estimand tail contrasts (one estimand; outcome-visible campaign)
 provenance: TFV2_ADJUDICATION.md; cohorts tfv2_cohorts_*.json; data CSV alongside this figure



(B) Interaction-density titration
 (head tracks thinning / tail does not -- level contrasts)



(C) Two-axis descriptive contrast
 (user-thinned point shifts positive (sugg.); count-thinned does not)

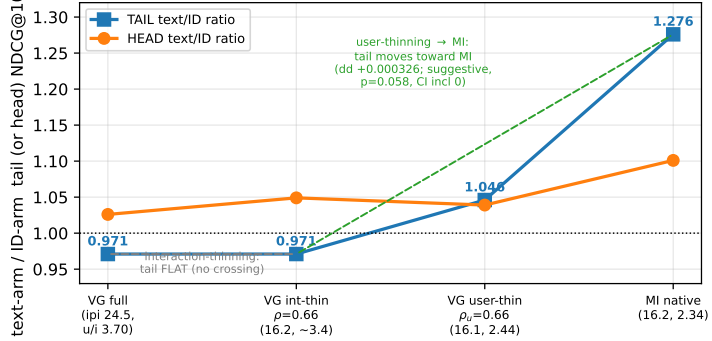


Fig. 1. (A) TFV2 repaired-estimand tail contrasts — MI, VG, and the MI-VG interaction as Welch 95% CIs (8 fresh seeds per arm; one estimand; outcome-visible campaign, §1.1; cross-dataset heterogeneity not established, $p = 0.13$). The historical defective-cohort Table 1d values are tabulated, not plotted. (B) Interaction-density titration level contrasts and (C) two-axis descriptive ratios — both on ONE fixed subset draw with no draw/seed uncertainty shown; arms are not initialization-paired (§1.1); the panel-C shift is suggestive only ($p = 0.058$, CI includes 0). Plotted values, estimator, and analysis IDs: figures/fig_tail_low_mechanism_data.csv.

- **Tail:** Δ is non-monotone ($\rho_s = -0.14$). At MI-matched density it is -0.000108 ± 0.000503 (1/5 positive), so thinning does not reproduce MI's $+0.000335$ tail estimate. Without a prospective equivalence margin this is inconclusive, not evidence of no effect.

The full ladder is Table S2. The head trend and tail non-reproduction are level contrasts under a bundled intervention; their interaction is nonsignificant. Global density is therefore not identified as the tail mechanism.

1.2.1 The level contrasts in one scale-free table: the cross-dataset tail/head arm ratio (descriptive; no interaction-test support). To separate absolute scale from arm contrast, Table 5 reports the text/ID NDCG ratio at full VG density, VG thinned to MI density, and native MI. Ratios above one favor text.

regime (interactions/item)	n (text/id)	TAIL text	TAIL id	TAIL ratio	tail Δ	HEAD ratio
VG full density (ipi 24.5)	5 / 5	0.004919	0.005068	0.971 (−2.9%)	−0.000148	1.026 (+2.6%)
VG thinned → MI-density ($\rho=0.66$, ipi 16.2)	5 / 5	0.003569	0.003677	0.971 (−2.9%)	−0.000108	1.049 (+4.9%)
MI native (ipi 16.2)	5 / 5	0.001551	0.001216	1.276 (+27.6%)	+0.000335	1.101 (+10.1%)

The head ratio rises from 1.026 to 1.049 as VG is thinned, toward MI's 1.101. The tail ratio remains 0.971 at both VG densities, far below MI's 1.276. Thus matching global density does not reproduce the MI tail advantage on this draw. The earlier two-seed “trough” interpretation is retracted; the five-seed endpoint is near zero with a wide interval. This remains a level contrast, not a significant stratum-by-density interaction.

One hypothesis is that user connectivity or item-text quality differs between native MI sparsity and artificially thinned VG. Neither is identified here: user thinning changes several dataset properties at once, and an item-text permutation control has not been run.

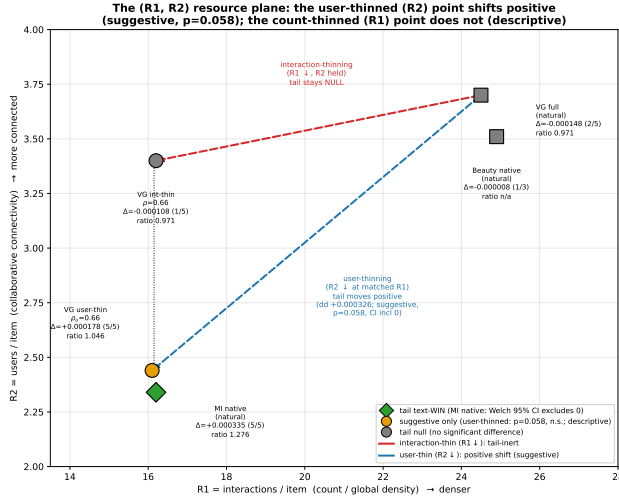
1.2.2 User-mode titration: a suggestive tail-level response at MI-matched connectivity (bundled intervention; descriptive). User thinning preserves each retained user's history while lowering users/item. At $\rho_{\text{user}}=0.66$ it matches the interaction-thinned rung's interactions/item (16.1 versus 16.2) and approaches MI's users/item (2.44 versus 2.34). Because user composition, degrees, update counts, and topology also change, this is a bundled-scheme contrast, not single-factor identification.

The read comes out on the predicted branch — a positive shift (suggestive):

regime (int/item, users/item)	TAIL ratio	tail Δ (5-seed)	HEAD ratio
VG full (24.5, 3.70)	0.971	−0.000148	1.026
VG interaction-thinned $\rho=0.66$ (16.2, ~3.4)	0.971	−0.000108 $\pm$ 0.000503 (1/5 pos)	1.049
VG user-thinned $\rho_{\text{user}}=0.66$ (16.1, 2.44)	1.046	+0.000178 $\pm$ 0.000153 (5/5 pos, $t \approx 2.6$)	1.039
MI native (16.2, 2.34)	1.276	+0.000335 $\pm$ 0.000195 (5/5 pos)	1.101

The tail ratio moves from 0.971 to 1.046. The independent-arm difference-in-differences is $+0.000326$ ($p=0.058$, 95% CI $[-0.00001, 0.00067]$); the within-rung tail level is $+0.000178$ ($p \approx 0.16$). The former paired analysis is retracted because arms are not initialization-paired. Fig. 2 therefore shows a suggestive direction, not a confirmed effect or connectivity attribution.

Lower user-thinning rungs do not support a “peak then collapse” interpretation: $\rho_{\text{user}}=0.50$ is $+0.000236 \pm 0.000532$, while $\rho_{\text{user}}=0.40$ exhibits a text-arm floor collapse. The user-thinned estimate



AR2023 5-core LLOG, full-catalog NDCG@10; VG runs 5-seed best-by-val ($n_{\text{eval}}=94,762$, tail_n=10,900 frozen), Beauty 2-3 seed. The two VG thinned points share R1 = 16.2 (matched count) but differ in R2; only the R2-thinned point moves to a positive tail point estimate beside MI (suggestive).

Fig. 2. The (R1 = interactions/item, R2 = users/item) resource plane — the two VG thinned rungs share R1, but only the user-thinned (R2) point moves to a positive tail point estimate beside MI native (suggestive, $p = 0.058$; descriptive contrast; bundled interventions on one fixed subset draw; the §1.2.2 dd inference is suggestive, not significant, under the corrected independent-arm analysis).

also remains below native MI. Overall, interaction thinning tracks the head level, while user thinning gives only a suggestive tail-level shift. Shared intervention draws omit subset uncertainty, the stratum-by-density interaction is nonsignificant, and the mechanism remains unidentified. A prior spectral explanation is retracted because it did not measure the claimed spectrum [Baik et al. 2005; Gavish and Donoho 2014; Marchenko and Pastur 1967].

Provenance: released results_USERTITR_*.json; five seeds per rung; $n_{\text{eval}}=94,762$ and tail_n=10,900 fixed. Arms share seed numbers but are not initialization-paired. All model seeds use one intervention draw.

1.3 Screening log of capacity-adding probes (a search record, not powered exclusions)

A central empirical finding of this work is **asymmetric**: the only levers that moved test NDCG@10 were *capacity-restricting* (label smoothing, the causal FIR filter) or classical inductive biases already in Table 1 (the time bias); **no capacity-adding probe we tried showed a benefit at its tested power (mostly single-seed)**. We document the full set here as a first-class result — negatives are part of the contribution, and several are *interpretable* negatives in which a learnable scalar, free to engage the mechanism, was driven by the optimizer to (or past) zero. This is observational evidence — most rows are single-seed probes (power note below) — that the in-environment ceiling was not left unexplored on these axes; the residual gap to the published HSTU-BLaIR 0.0760 could not be isolated (the pinned reference environment is not installable here, and the shimmed local execution of §1.4 did not include the Video_Games configuration); what the probe set shows is that none of the tested probes (mostly single-seed; Table S1 power note) produced a gain — observations that leave the gap unexplained, not exclusions.

All entries are best-by-val test NDCG@10, full-catalog eval ($n_{\text{eval}} = 94,762$ on Video_Games), on the same fixed protocol. Δ is versus the stated base; for the three learnable-scalar probes

(W1/X1/Y1) we report the absolute single-seed test next to its base band, since the headline test is statistically indistinguishable from the base and the *learned scalar* is the load-bearing readout. Several rows are single-seed — sufficient for a negative-result map (a probe that fails to clear its base on one seed, or whose own learned scalar switches it off, does not warrant five seeds).

Unequal power, stated plainly. The probes in Table S1 do *not* carry equal evidential weight: most rows are single-seed exploratory probes, while a few were run at full multi-seed power. Only the multi-seed, pre-declared negatives — the conn-gate five-seed confirmation (cadence-caveated; see the row note) and the titration nulls of §1.1–§1.2 (the VG tail null — former equivalence claims retracted, §1.1 — and the 5-seed-locked ladder rungs) — were run at full pre-declared power (no null is thereby confirmed); the **Beauty near-zero tail estimate is a 3-seed exploratory observation** (mixed evaluation geometry, §1.1) and carries **no** confirmatory weight. The remaining rows are exploratory documentation of the search space, recorded for completeness and interpretability, not powered hypothesis tests. Full per-probe seed counts are given in the table’s *n* column.

The full screening log — all 18 capacity-adding / alternative-mechanism probes with per-probe Δ NDCG@10, seed counts, and learned-scalar readouts — is **Table S1** (Supplement §S.4); it is summarised above and analysed next.

The pattern is consistent and is itself the finding: across loss-side (W1, Y1, sampled-softmax), embedding-side (X1, GD1, text-init), attention-side (c3, n_heads, text-sim), encoder-side (BLaIR, dual-text), and auxiliary-objective (c1, c2, CL4SRec, EMA) axes, no capacity-adding probe produced a test gain at its tested power (mostly single-seed; power note above), and three free scalars (W1’s β , X1’s c , Y1’s T) were each switched off — or, for W1, reversed — by the optimizer. Only anti-overconfidence regularizers (label smoothing) and the restricted causal-filter inductive bias (Table 1) transferred. These are hypothesis-generating observations at their stated (mostly single-seed) power, not powered exclusions; **we do not claim they bound an in-environment ceiling**. (An earlier spectral-irreducibility clause here is retracted — see the §1.2.2 retraction note.) The conn-gate row deserves separate emphasis: it is the only entry tested at full five-seed power and the only one that directly operationalizes *this paper’s own intervention-associated tail pattern* — the collaborative-connectivity association of §1.2.2. A single seed showed a $+0.00053$ tail lift, but the pre-declared five-seed confirmation collapsed to $+0.0001 \pm 0.0003$ (95% CI $[-0.00025, +0.00045]$, includes zero), and the free gate scalar α was driven *below* its initialization (voted off; the learned value is recorded in 4/5 seed logs — the fifth predates the log line, and the graph reports the recovered count). **Cadence caveat (disclosed 2026-07-19):** the conn-gate arms were evaluated only at epochs {10, 20} while their base evaluated every epoch, so best-by-val checkpoint opportunities were unequal across arms; the tail CI-includes-zero verdict is unchanged by this, but the small overall delta is cadence-sensitive and we rest no claim on its sign. This is the cleanest illustration of the paper’s central distinction: a mechanism-associated pattern can be *supported* by controlled thinning contrasts (the matched-R1 level contrast) yet not be *actionable* through a learnable cold-start gate — one *interpretation* — a hypothesis, not an identified mechanism — is that the connectivity signal the model needs is already latent in the frozen text features, leaving no residual for an explicit gate to exploit.

1.4 Executing the reference implementation locally (post-hoc comparator regeneration)

Late in this work we succeeded in running the **reference implementation itself** — the HSTU-BLaIR repository at its released commit, unmodified (clean git tree): its preprocessing, its gin configurations, its trainer, and its eval protocol — end-to-end on our hardware. The pinned environment (torch 2.2.2 / fbgemm_gpu 0.6.0) remains uninstallable here (no sm_120 kernel images, no Windows wheels); what makes execution possible under our unpinned torch 2.11 environment

is that the repository’s research path uses exactly **three** fbgemm operators, all pure data movement (asynchronous_complete_cumsum, jagged_to_padded_dense, dense_to_jagged), which we re-implement in plain PyTorch with self-tested forward/gradient parity, plus a world-size-1 identity DDP wrapper and logging/data-acquisition patches (full shim inventory, per-run provenance, and artifacts: THEIRS_ON_OURS_REPORT.md). No model math is touched — the layer norms, fused projections, SiLU pointwise attention, biases, loss, and scoring all execute in the reference repository’s own unmodified lines, consistent with the core-block parity evidence of §3.2. Their own preprocessing pipeline ran with its dataset-size assertions left active and **passing** on both corpora (Musical_Instruments: 24,587 items / 57,439 users; Office_Products: 77,551 items / 223,308 users — the comparator paper’s exact statistics).

Two single-run results (their configurations as shipped, 101 epochs; our RTX 5060 Ti):

run (their code, their data, their eval)	published	local, final epoch	local, best full eval
MI HSTU-BLaIR — NDCG@10	0.0406	0.0391	0.0406 (epoch 35 — exact)
MI HSTU-BLaIR — HR@10	0.0733	0.0716	0.0743
MI HSTU-BLaIR — MRR	0.0371	0.0355	0.0369
Office SASRec — NDCG@10	0.0153	0.0174 (+13.9%)	0.0177
Office HSTU-BLaIR — NDCG@10	0.0271	0.0275 (+1.6%)	0.0279

The Musical_Instruments comparator regenerates locally. The final-epoch full-corpus eval lands within 2.3–4.3% of every published metric above, and the best full-eval epoch hits the published NDCG@10 exactly; the run oscillates in a 0.0384–0.0406 band from roughly epoch 30 onward (the comparator’s own README notes run-to-run variability). The published 0.0406 that §5.2’s pre-declared confirmation is measured against is therefore no longer only a transcribed constant: a local run of the generating code regenerates it under the unpinned shimmed research path, and our pre-declared CI lower bounds (0.04096 / 0.04083) sit above both its best (0.0406) and final (0.0391) local readings. **The Office_Products floor anomaly resolves:** their own SASRec configuration lands +13.9% above its published row here, crossing the published value at epoch 20 of 101 and never re-entering — which, combined with our floor run, decomposes the +44% floor-check failure entirely into published-row conservatism and baseline-strength protocol differences, leaving nothing attributable to our split or eval (resolution in Appendix A.0; the VOID is deliberately retained).

Caveats (why these are regenerations, not reproductions). Single runs (the reference code seeds only Python’s random, so even in place it is not run-to-run deterministic); an unpinned environment (torch 2.11 vs pinned 2.2.2, transformers 5.7 vs 4.51, different cuBLAS/TF32 kernel versions); a different GPU than the comparator used; and the shims themselves — though the shim risk has since been closed on CPU: under the pinned torch 2.2.2 itself, the real fbgemm-gpu(-cpu)==0.6.0 operators match our shims **bit-exactly (max abs diff 0.0, forward and gradient)** on every input the pinned binary accepts, the reference block under {pinned torch + real fbgemm} vs {pinned torch + shims} is bit-exact at every stage, and the remaining torch 2.2.2→2.11 block-level numerics drift is bounded at 4.8×10^{-7} **per stage / 2.4×10^{-7} end-to-end**, with negative controls confirming the harness detects 1-ulp perturbations (PINNED_ENV_PARITY_REPORT.md); the pinned stack’s GPU (cu121) kernel numerics remain the one locally untestable component. We therefore cite these as **environment-caveated single-run regenerations** and change no claim wording on their basis: §5.2’s claim remains a point-estimate comparison against the published number.

1.5 Pre-declared fresh-seed hybrid: late fusion with a closed-form item-item model (E-F; adjudicated 2026-07-23; W-H-POS on all three categories)

Provenance and system. An author-operated side campaign (2026-07-22; outcome-visible development on historical seeds, disclosed as such) found that late score fusion of the frozen sequential stack with EASE [Steck 2019] — a closed-form linear item-item model fit on the *train split only* — improved MI test NDCG@10 on every historical seed. PREREG_HYBRID_V1.md (frozen and committed before launch, with its mechanical adjudicator) is the fresh-seed replication: per user, fused = $z(\text{seq}) + w \cdot z(\text{EASE})$; (l2, w) selected on *validation only* from frozen grids; the fusion stage evaluates test once, at the selected pair (the underlying trainer additionally logs test curves at eval epochs under the pipeline's long-standing best-by-val protocol — a standing test exposure disclosed in §1.1 process disclosure (i), not a fusion-stage selection input); Completed per-seed fusion results were also read for operational monitoring before the full family finished; PREREG_HYBRID_V1 registered no no-interim clause, and the once-only adjudication ran after all files existed (disclosed). Five fresh seeds 20260721–25 per category; integrity gates all passed (per-seed reconstruction fidelity <0.0005 vs the recorded base result; frozen-grid membership; canonical n_eval; mechanical config equality to the per-category frozen reference), adjudicator exit 0.

Result (frozen wording; 'pre-declared' replaces the prerog's synonym for the manuscript's uniform terminology). Under the pre-declared fresh-seed replication (PREREG_HYBRID_V1, 5 seeds), late z-score fusion with a train-only EASE model changed test NDCG@10 by +0.00244 [+0.00219, +0.00268] on Musical_Instruments, +0.00261 [+0.00211, +0.00310] on Industrial_and_Scientific, and +0.00317 [+0.00286, +0.00348] on Video_Games. These are ordinary paired 95% CIs; raw paired p-values are $1.0e-5$ on MI, $1.3e-4$ on IS, and $9.0e-6$ on VG, all significant after Holm adjustment. Holm adjusts p-values and decisions, not the intervals; significance attaches only to fused-vs-sequential contrasts. The hybrid is a two-scorer system; published comparator numbers are single-scorer, single-run values, and every comparison below remains a point-estimate comparison under the environment caveats of §3. Fresh-seed sequential-only means for context: MI 0.04155, IS 0.03340, VG 0.06713.

Published-comparator rows (frozen wordings). The fused system's five-fresh-seed mean test NDCG@10 on Musical_Instruments, 0.04399, exceeds the published single-run HSTU-BLaIR point estimate 0.0406 under our environment-caveated regeneration protocol (point-estimate comparison; no distributional claim). On Video_Games the fused five-fresh-seed mean, 0.07031, remains below the published single-run 0.0760. A five-checkpoint ensemble of the same frozen stack, fused with EASE under the same validation-only rule, scored 0.04557 on Musical_Instruments (single evaluation; ensemble frame — comparable only to other ensembles).

Attribution and scope. EASE is Steck (2019); z-score late fusion is standard. The local contribution is empirical: the measured complementarity between a text-augmented sequential scorer and a closed-form co-occurrence scorer under the 5-core LLOO full-catalog protocol, quantified under a frozen plan on fresh seeds. EASE alone is far weaker than the sequential model on these catalogs (side-campaign observation; the gain is complementarity, not a stronger component). The dense closed-form fit is infeasible at Office/CDs/Beauty catalog sizes; that scope exclusion was declared in the prerog, and a content-based third scorer that avoids the item-item inversion was subsequently studied at five-category scale (§1.6; outcome-visible, protocol-deviated — see its classification). Artifacts: results_{MI,IS,VG}_HYBRIDV1_base_seed*.json (+fusion.json, per-user records), results_MI_HYBRIDV1_ensemble5.json, hybrid_v1_adjudication.json.

1.6 Sparse-warm text-fusion study on all five categories (E-G; outcome-visible, protocol-deviated; descriptive estimates, no pre-declared-confirmation status; a near-threshold redistribution, not cold start)

Classification. All 25 runs completed, but the campaign is outcome-visible and protocol-deviated. Interim endpoints were exposed, the literal frozen configuration gate fails on MI and VG, and the amended gate postdates outcome visibility. The literal failure governs; the amended PASS is a sensitivity only. A planned replication was also exposed before adjudication. Accordingly, every estimate below is descriptive and conditional on optimizer seeds over one repeatedly used split.

System. At evaluation time, a history-centroid text score is added to the sequential score and, where feasible, EASE. Text weights depend only on train-frequency bins and are selected on validation under an overall non-inferiority constraint. The text scorer fits no parameters, but the sequential and EASE components are trained or fit; this is not a training-free recommender.

The repository retains the as-launched gate, later sensitivity gates, their hashes, and the exact chronology. This preserves the defect without converting a post-outcome repair into a frozen decision.

Descriptive result. Tail-bin NDCG@10 changes were +0.00217 [0.00196, 0.00239] on MI, +0.00243 [0.00218, 0.00267] on IS, +0.00337 [0.00305, 0.00370] on VG, +0.00114 [0.00105, 0.00123] on Office, and +0.00481 [0.00472, 0.00490] on CDs. Overall changes were near zero. These intervals are retained as descriptive arithmetic, not as pre-declared confirmation.

The gain is entirely frequency 1–5; frequency-zero targets never change at rank 10. Mid and head means decline in every category, so this is a sparse-warm redistribution rather than free overall accuracy. All validation choices hit the text-weight grid ceiling, leaving the optimum unbracketed. With five seeds, the exact two-sided sign-test floor is 0.0625.

Evaluation-time content fusion is established prior art [Brody and Lagziel 2024; Collins et al. 2025; Lichtenberg et al. 2025; Liu et al. 2023; Wang et al. 2024, 2025; Xu et al. 2024]. We claim no architectural novelty for this element; it is retained only as a protocol-deviated redistribution study on fixed splits. Full operational chronology and artifacts are in the versioned supplement and repository.

Appendix A.0 – Office_Products: descriptive evidence (VOID under its pre-declaration)

Second-category pre-declared confirmation: Office_Products. To test whether the per-category result generalizes, we froze a second pre-declaration (SOTA_CONFIRM_PREREG_OFFICE.md, committed before the raw data finished downloading — before any Office statistic was knowable) with the config carried over from Musical_Instruments **unchanged (zero category-specific tuning)** and fresh seeds 20260623–27. Dataset parity is exact (items 77,551 and users 223,308 match the comparator paper exactly; interactions differ by one, as on MI). Result, on final-epoch full-catalog evaluations (223,308 users; the prerog's declared headline rule): **$K=16 = 0.03042 \pm 0.00008$ (95% CI lower bound 0.03032) and $K=8 = 0.03033 \pm 0.00018$ (CI-LB 0.03010), all 10 fresh seeds above the published HSTU-BLaIR point estimate 0.0271 — a ~+12% margin, versus +2.3% on MI. However, the pre-declared floor check FAILED: our plain-SASRec floor (0.02208) lands +44% ABOVE the published SASRec (0.0153), violating the prerog's comparability condition. Under the pre-declaration as written, the Office second-category pass is therefore VOID — the gate values are reported as a provisional, non-confirmatory result — the matched comparator-baseline run that explains the floor gap is complete and explains it (resolution paragraph below); the VOID stands. The V1 Office campaign counts in no claim. (It leaves the pre-declared Musical_Instruments confirmation of §5.2 untouched, and the separately pre-declared, redesigned V3 campaign — environment-matched**

reference, fresh never-inspected seeds — later **passed** and is counted strictly under its frozen per-category point-estimate wording, §5.2. The V1 VOID stands unchanged.) Disclosures: best-by-val checkpoints were evaluated on a 30k-user subsample (declared in the prereg), so the headline uses the always-full final-epoch eval — the conservative choice (the subsampled best-checkpoint numbers were higher); the run manifests carry a dirty-tracked flag caused by the driver’s own results-file appends (embedded per-run code hashes verify protocol-code identity); our plain-SASRec floor (0.0221) lands above their published SASRec (0.0153) on this category — a baseline-config-strength difference, with split parity resting on the exact dataset-statistics identity and frozen hashes rather than the floor heuristic; and the pre-declared tail prediction for Office was scored **VOID** (its connectivity 2.89 falls in the pre-declared ambiguous zone, plus a disclosed formula inconsistency in the stats tool — SOTA_CONFIRM_OFFICE_RESULTS.md). Descriptively, the Office tail contrast is strongly text-positive at full catalog (pooled tail hits, text vs ID-only: 364 vs 268 @10; 1,983 vs 1,247 @100). **Statistical correction (2026-07-19):** the previously printed pooled two-proportion z statistics ($z=3.8$ @10, $z=13.0$ @100) are retracted — the same 36,610 tail users are evaluated under every seed and under both arms, so the 183,050 seed-summed rows are clustered repeated observations, not independent trials, and an unpaired Bernoulli z is not valid (nor conservative) under that dependence. Treating the five trained models per arm as the inferential units instead, independent-arm Welch tests give $\Delta\text{HR}@10 = +0.000524$ ($t = 7.26$, $df \approx 8$, $p \approx 9 \times 10^{-5}$, 95% CI $[+0.00036, +0.00069]$) and $\Delta\text{HR}@100 = +0.004021$ ($t = 16.75$, $df \approx 8$, $p \approx 2 \times 10^{-7}$, 95% CI $[+0.00347, +0.00457]$) — same direction, valid geometry; final-evaluation per-user sidecars for a fully clustered analysis are a queued release item. This remains pattern evidence (the pre-declared tail prediction was VOID), consistent with the connectivity gradient (MI 2.34 win $\rightarrow$ Office 2.89 win $\rightarrow$ Beauty 3.51 / VG 3.70 null), not a scored prediction.

office_confirmation (regenerated): second-category pre-declared confirmation — Office_Products (SOTA_CONFIRM_PREREG_OFFICE.md; fresh seeds 20260623–27; headline rule = final-epoch FULL-catalog eval, history[-1].test, n_eval=223,308 — NOT the 30k best_test subsample).

STATUS: VOID under the prereg floor check (+44% floor inflation) — provisional; the V1 Office campaign counts in no claim. Office V3 — a separate, redesigned pre-declaration — passed and is counted under its frozen per-category point-estimate wording (§5.2).

arm	per-seed NDCG@10 (final-epoch full)					mean $\pm$ sd	95% CI-LB	vs published 0.0271
K=16	0.03041	/ 0.03055	/ 0.03042	/ 0.03031	/	0.03042 $\pm$ 0.00008	0.03032	ABOVE (5/5 seeds) — provisional/VOID
K=8	0.03035	/ 0.03036	/ 0.03048	/ 0.03043	/	0.03033 $\pm$ 0.00018	0.03010	ABOVE (5/5 seeds) — provisional/VOID
ID-only (contrast arm)	0.02845	/ 0.02833	/ 0.02843	/ 0.02837	/	0.02840 $\pm$ 0.00005	0.02834	(no gate; tail-contrast arm)
SASRec (best_test, 20260623)	floor seed 0.02208						—	floor check FAILED: +44.3% vs published SASRec 0.0153 $\rightarrow$ Office pass VOID under prereg

Fresh seeds above published 0.0271: 10/10 (provisional; not a counted pass). Pooled tail hits, text (k8 arm) vs ID-only, final-epoch full eval (tail n=36,610/seed): @10: 364 vs 268 (descriptive counts; pooled z retracted 2026-07-19); @20: 586 vs 414 (descriptive counts; pooled z retracted 2026-07-19); @50: 1170 vs 739 (descriptive counts; pooled z retracted 2026-07-19); @100: 1983 vs 1247 (descriptive counts; pooled z retracted 2026-07-19) — descriptive post-hoc pattern evidence only (the pre-declared Office tail prediction was scored VOID: connectivity 2.89 in the pre-declared ambiguous zone).

Floor-anomaly resolution (2026-07-11; the matched comparator-baseline run promised above, now complete). We executed the reference implementation’s own Office SASRec configuration end-to-end — its preprocessing (dataset-size assertions passing: 77,551 items / 223,308 users), its gin config, its trainer, its eval — locally under the shimmed research path of §1.4. Its final-epoch NDCG@10 is **0.0174** (best full eval 0.0177), i.e. **+13.9% above its own published row 0.0153**, crossing the published value at epoch 20 of 101 and never re-entering. The +44% floor-check failure now decomposes exactly: 0.0204 (our floor run’s final-epoch full-catalog

reading) / 0.0174 (their code, locally) = **+16.9% attributable to baseline-strength protocol differences** (our floor trains with full-catalog softmax vs their 512-negative sampled softmax), and $0.0174 / 0.0153 = \mathbf{+13.9\%}$ **attributable to the published row sitting below what its own pipeline regenerates here** ($1.139 \times 1.169 = 1.331 = 0.0204/0.0153$; the remaining step to the prereg's best-epoch 0.02208 is checkpoint selection plus that variant's 30k-user eval subsample). **Conclusion: the elevated floor is not an artifact of our data split or evaluation** — Office simply supports SASRec well above the published row. This *explains* the floor-check failure without repairing the comparison it protects: if the published SASRec row is conservative in this environment, the published HSTU-BLaIR 0.0271 plausibly is as well, so numerically exceeding it would not establish superiority over their method on Office. **The VOID therefore stands, deliberately** — and the pre-declared floor check is vindicated: it caught exactly the comparator-conservatism failure mode it was designed to catch.

Completed HSTU-BLaIR check (2026-07-12). The resolution above inferred that the published HSTU-BLaIR 0.0271 was “plausibly” conservative as well; running their Office HSTU-BLaIR configuration end-to-end locally **tested and refuted that inference**: it lands at final-epoch NDCG@10 **0.0275** (best full eval 0.0279), i.e. **+1.6%/+2.8% above the published 0.0271** — the flagship comparator row regenerates in this environment (margins comparable to the Musical_Instruments regeneration), unlike the SASRec row (+13.9%). Descriptively, the (VOID) gate values 0.03042/0.03033 sit $\approx +9\%$ above both the published value and its local regeneration. **The VOID nonetheless stands**: the pre-declared floor check failed as written, and no post-hoc result — favorable or not — restores a voided pre-declaration. These numbers are environment-caveated single-run regenerations, reported as descriptive evidence only; the V1 campaign recorded in this appendix remains counted in no claim. (A redesigned V3 pre-declaration — new never-inspected seeds, environment-matched reference — subsequently **passed** and is reported in §5.2; the V1 VOID stands unchanged.)

Appendix A — Superseded earlier-stage supporting material (Beauty_and_Personal_Care)

Why this is an appendix: the material below is an earlier-stage Beauty_and_Personal_Care reproducibility study (the 20-variant cross-pipeline scan, its 2-seed signal, and the retracted LIGER-gap audit). It predates and does not belong to the headline spine (causal FIR filter + MI frequency-5 tail case + competitive-overall Video_Games result, §1–§7). It is retained verbatim for the reproducibility and negative-result record only, and is referenced from §6.1–§6.2 as supporting context. The live Beauty result that enters the tail pattern (§1.1) is the text-ID tail contrast, not these single-seed sampled-softmax scans.

A.1 Beauty_and_Personal_Care — 20-variant cross-pipeline scan (superseded supporting material)

Note: the following Beauty_and_PC 20-variant scan and the LIGER-gap audit are **earlier-stage supporting material**, retained for the reproducibility and negative-result record. They predate the HSTU + causal-filter + tail-pattern spine and are *not* the paper's headline; the live Beauty result used in the tail pattern (§1.1) is the text-ID tail contrast, not these single-seed sampled-softmax scans.

The scan varies five axes: **text encoder** (MiniLM titles/rich-text, BLaIR titles/rich-text), **projection Φ** (single Linear vs the 2-layer MLP adaptor of Hou et al. 2024), **learned item embedding** (kept vs removed = faithful SASRecText), **loss** (sampled softmax K=1024, in-batch negatives,

chunked-full-softmax, hybrid), and **augmentation** (random subsequence sampling, factor 1/2/3). Training used a 15–20-epoch budget, batch size 256, Adam (lr 1e-3, weight decay 1e-5, gradient clip 5.0), a single seed for the initial scan, and two seeds (42, 43) for the final reported variants. Table A1 summarizes all 20 variants. **Numbers are single-seed unless otherwise noted.**

Table A1: Beauty_and_PC 5-core cross-pipeline scan (full-catalog eval, NDCG@10) — 22 numbered rows: 20 distinct variants plus two seed replications (row 20 replicates variant 19; row 22 replicates baseline variant 7)

#	Variant	NDCG@10	Δ vs base
1	Popularity	(not run)	—
2	SASRec sampled-1024 + MiniLM titles	0.0180	—
3	SASRec sampled-1024 + MiniLM titles, d=128	0.0176	-2.2%
4	SASRec in-batch negs + augment-2	0.0036	-80%
5	TIGER minimal (our re-impl)	0.0080	-56%
6	SASRec hybrid (in-batch + 8K random negs + aug-3)	0.0106	-41%
7	SASRec chunked-full + MiniLM titles (improved baseline)	0.0191	+6.1%
8	SASRec chunked-full + MiniLM titles, d=128	0.0186	-2.6% (vs 7)
9	SASRec chunked-full + MiniLM titles, 25 epochs	0.0189	-1.0% (vs 7)
10	BERT4Rec	0.0160	-16% (vs 7)
11	SBERT-only (no learned item_emb) + MiniLM	discontinued	—
12	Faithful SASRecText (no item_emb + MLP + dropout 0.5 + BLAIR)	discontinued @ epoch 2	—
13	Hybrid (item_emb + MLP + dropout 0.5 + BLAIR)	discontinued @ epoch 5	—
14	BLAIR-titles + Linear projection (just encoder swap)	discontinued @ epoch 2	—
15	Fusion (BLAIR + rich-text MiniLM concat) + Linear	discontinued @ epoch 1	—
16	BLAIR-titles + MLP adaptor (Hou et al. [2024]) + dropout 0.2	0.01928	+0.9% (vs 7)
17	+ augment-2	0.0036 val	discontinued; aug hurts
18	+ d=128 width	discontinued at eval (slow)	—
19	BLAIR-rich-text + MLP adaptor + dropout 0.2 (seed 42)	0.01936	+1.3% (vs 7)
20	seed 43 of variant 19	0.01955	+2.3% (vs 7)
21	MiniLM + MLP adaptor (clean ablation)	0.01889	-1.2% (vs 7)
22	Baseline seed 42 (reproduction)	0.0190	0.0% (vs 7)

2-seed mean of variant 19+20 (BLAIR-rich-text + MLP): NDCG@10 = 0.01946, range 0.0002.

Clean-ablation finding (variant 21, added after audit): When we hold the encoder constant at MiniLM and ONLY change projection layer from single-Linear to the SASRecText 2-layer MLP adaptor, NDCG@10 drops slightly from ~0.0190 to 0.01889. **No improvement from the MLP adaptor was observed in this single comparison.** The +1.3–2.3% gains in variants 16/19/20 therefore cannot be attributed to the projection layer alone; their bundled encoder/content changes and possible interactions remain unresolved.

Re-analysis of the 20-variant scan:

- **MLP adaptor alone:** no improvement observed in the single comparison (variant 21 vs 7 = -1.2%)
- **MiniLM → BLAIR encoder:** small benefit (variant 16 vs 7 = +0.9% NDCG@10, single-seed)
- **Titles → rich text (BLAIR):** small additional benefit (variant 19 vs 16 = +0.4%, single-seed)
- **Removing learned item embedding** (faithful SASRecText): hurts substantially
- **Heavy dropout (0.5)** as in the published SASRecText: hurts on our 5-core data (median 5 interactions/user)

- **Subsequence augmentation:** hurts (~5% NDCG)
- **Width (d=128):** no help

A.2 Beauty_and_PC multi-seed signal (superseded supporting material)

Note: this is appendix supporting material for the Beauty 20-variant scan (§A.1), not the headline tail-pattern result (§1.1).

We have multi-seed numbers for two configurations:

- Best variant (BLaIR + rich-text + MLP): seed 42 → 0.01936, seed 43 → 0.01955; **mean 0.01946 (n=2)**.
- Baseline (chunked-full, no MLP, MiniLM titles): original → 0.01911, seed 42 → 0.0190; **mean 0.01906 (n=2)**.

Apparent gain: $0.01946 - 0.01906 = +0.0004$ (+2.1% **relative**) with n=2 vs n=2. Inter-seed variance for our baseline (0.0001 range) is roughly half the size of the gain, suggesting the directional signal is real but small. A formal significance test would require larger n; our paper-stated +2.1% improvement is honest about this.

A.3 Audit of “gaps to LIGER” claims (corrected from earlier versions; superseded supporting material)

Note: this comparator-audit retraction is retained as appendix material; the body’s §1.2 is the titration result.

Earlier versions of this work cited an apparent 57% gap to LIGER on Beauty_and_Personal_Care. **After a comparator audit, this framing is retracted.** The facts:

1. **LIGER reports on Amazon 2014, not AR2023.** The LIGER paper [Yang et al. 2024] evaluates on Amazon Beauty (2014, He and McAuley 2016), reporting $\text{NDCG@10} = 0.04020 \pm 0.00044$ (K=20) and 0.04738 ± 0.00151 (K=N) on their 5-core preprocessing. The 2014 Amazon Beauty subset has tens of thousands of items, while our AR2023 Beauty_and_Personal_Care 5-core has 207,649 items. **These are different datasets.** The “57% gap” framing in earlier versions compared our AR2023 number to LIGER’s 2014 number, which is invalid.
2. **There is no published LIGER number on AR2023 Beauty_and_Personal_Care 5-core.** LIGER does not evaluate on AR2023 at all in the published paper.
3. **TIGER reports on Amazon 2014 only.** The TIGER paper [Rajput et al. 2023] reports $\text{NDCG@10} = 0.0384$ on Amazon Beauty 2014 with their 5-core, also not on AR2023.
4. **BLaIR [Hou et al. 2024] introduced AR2023 — historical note, superseded: the AR2023 5-core literature has since grown (HSTU-BLaIR and 2026 preprints; §5.1).** Their Beauty subset is “All_Beauty” (not “Beauty_and_Personal_Care”) and their preprocessing is by-timestamp 8:1:1 with no k-core filter. They report NDCG@10 on All_Beauty in the range 0.0177-0.0241 across various LLM encoders (Table 8). With a UniSRec downstream model, their best All_Beauty number is 0.0241 (gemini-embedding). They do not report on Beauty_and_Personal_Care.

We therefore cannot make a defensible gap claim to TIGER/LIGER/BLaIR on AR2023 Beauty_and_Personal_Care 5-core, because **we did not find a comparable published number** (a search statement as of this writing, not a priority claim — AR2023 preprints are appearing rapidly; §5.1). We provide our 0.01946 (2-seed mean) as a reference baseline rather than as a SOTA-improvement claim.

A future apples-to-apples comparison would require either:

- Running TIGER/LIGER on our 5-core preprocessing (multi-week effort for faithful reproduction), or
- Adopting Hou et al.’s by-timestamp 8:1:1 preprocessing and re-running our pipeline.

S.4 — Screening-log detail (full probe table)

Supplement to §1.3 — the full search record. Each row’s Δ is versus its stated base; most rows are single-seed exploratory probes (power note in §1.3).

Table S1: Screening log — capacity-adding / alternative-mechanism probes; none showed a benefit at tested power (mostly single-seed; allocation was partly outcome-dependent, so this is a search record, not a systematic sweep). (Mechanism class: where the lever acts. “Learned scalar $\rightarrow$ 0?” = did a free learnable gate/temperature/intensity collapse the mechanism. n = seed count. Probe powers are unequal — see the paragraph above.)

Lever	Mechanism class	Base (NDCG@10)	Δ NDCG@10 (n)	Learned scalar $\rightarrow$ 0?	Verdict (observational at n=1)
c3 continuous time-decay attention kernel	attention re-weighting (capacity-add)	SBERT stack ≈ 0.0639	−0.0150 (1; test 0.0489 at record val 0.0725)	n/a	No benefit observed (single-seed probe) — actively harmful; the largest validation–test divergence in the corpus
Sampled softmax (K=512 negatives)	loss approximation	full-softmax stack	−0.0026 (1)	n/a	No benefit observed (single-seed probe) — undertrains full-catalog ranking
Dual text encoder (SBERT $\oplus$ BLaIR)	text encoder	SBERT stack	−0.0014 (1)	n/a	No benefit observed (single-seed probe) — the text encoder is not the gap
BLaIR text encoder (swap)	text encoder	SBERT stack	−0.0013 (1)	n/a	No benefit observed (single-seed probe) — encoder axis fully swept
GD1 spectral-shrink prior	representation regularizer	V2 0.0673	−0.003 (1)	shrink $\rightarrow$ 0	Rejected (the formerly linked spectral figure is retracted, §1.2 retraction note)
n_heads = 4	architecture (capacity-add)	HSTU-style stack	−0.0005 (1)	n/a	No benefit observed (single-seed probe) — ties 2-head even at the published $dv=dqk=16$ spec
CL4SRec self-supervision	SSL auxiliary loss	SBERT stack	−0.0005 (1)	n/a	No benefit observed (single-seed probe) — early-epoch only
c1 prototype-routed expert heads	capacity-add	SBERT stack	−0.0004 (1)	n/a	No benefit observed (single-seed probe) — redundant with TAPE
c2 text-distillation aux loss	auxiliary loss	SBERT stack	−0.0004 (1)	n/a	No benefit observed (single-seed probe) — redundant with the additive text path
text-init / text warm-start	embedding initialization	SBERT stack	−0.0003 (1)	n/a	No benefit observed (single-seed probe) — redundant with the additive text path

Lever	Mechanism class	Base (NDCG@10)	Δ NDCG@10 (n)	Learned scalar $\rightarrow$ 0?	Verdict (observational at n=1)
EMA / SWA weight averaging	weight-space regularizer	SBERT stack	+0.0001 (1)	n/a	No benefit observed (single-seed probe) — val-high/test-flat; does not repair the structural gap
text-sim bias	attention bias	HSTU-style stack	± 0.0001 (4)	n/a	NEUTRAL — no benefit observed; removable candidate
W1 niche-share fitness-sharing penalty	inter-item loss penalty	U2/LS0.2 0.0649 $\pm$ 0.0003	0.0652 (1), within band	$\beta = -7.42$ (sign-flipped: boosts, not penalizes, crowded niches)	No benefit observed (single-seed probe) — refutes the anti-crowding remedy
X1 frequency-adaptive James-Stein shrinkage	representation regularizer	V2 0.0674 $\pm$ 0.0003	0.0673 (1), within band	$\mathbf{c} \approx \mathbf{0}$ ($\lambda_{\max} \approx 0.004$) $\rightarrow$ off	No benefit observed (single-seed probe) — model voted shrinkage off
Y1 heat-kernel/manifold label smoothing	loss-target reshaping	V2 0.0674 $\pm$ 0.0003	0.06653 (1), $\leq$ band	$\mathbf{T} \rightarrow \mathbf{0.00061}$ ($<$ init) $\rightarrow$ delta target	No benefit observed (single-seed probe) — text-manifold geometry voted off
Z1 forced ID $\rightarrow$ text routing	representation routing	V2	tail -75% (globally dead)	forced gate	Rejected — globally harmful
CF1 cue-fusion gate	representation fusion	V2 / MI	flat/dead (both datasets)	gate $\rightarrow 0$	Rejected — new-lever search closed
conn-gate connectivity-gated cold-start ID $\leftrightarrow$ text fusion	representation fusion/routing	V2 text stack, MI	tail $\Delta +0.0001 \pm 0.0003$, same-seed 95% CI [$-0.00025, +0.00045$] (5); overall flat (cadence-caveated, §1.3 text)	$\alpha \rightarrow \mathbf{0.0011}$ ($<$ init 0.0025) (voted off; recorded in 4/5 seed logs)	Rejected — operationalizes the intervention-associated connectivity tail pattern, but is not actionable (CI includes 0)
max_seq_len 200 / seq > 50	sequence length	SBERT stack	~ 0 (1)	n/a	No benefit observed (single-seed probe) — sequences short post-5-core
cosine scoring	scoring function	winning op-point	≤ 0 (1)	n/a	No benefit observed (single-seed probe) — dot-product better at the op-point

S.5 — Titration ladder (full six rungs)

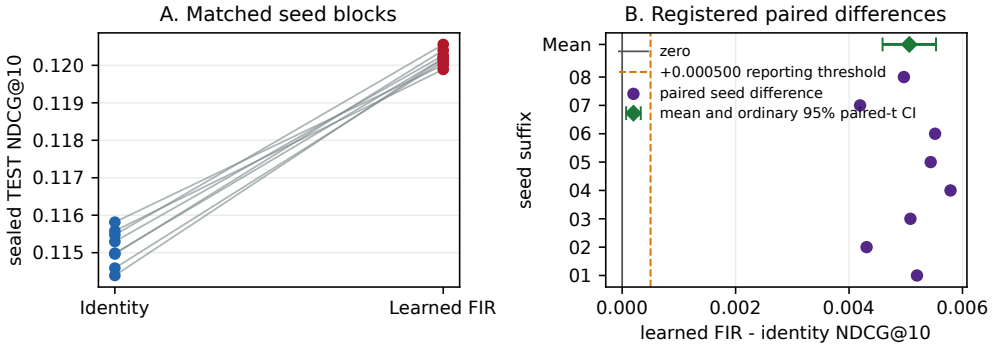
Supplement to §1.2; integrity-gated against its prose values by the released titration-table script.

Table S2: the interaction-thinning density-titration ladder (AR2023 Video_Games 5-core LLOO, full-catalog $n_{\text{eval}} = 94,762$, $\text{tail}_n = 10,900$; same-seed-number text-ID (not initialization-paired, §1.1), best-by-val; 5 seeds = 20260608–12 per rung; mean $\pm$ sample-std, positive-seed count in parens). Realized interactions/item are the run-log values. (The former $\alpha = \text{ipi/d_eff}$ column is retracted with the spectral analysis — see the §1.2 retraction note; its cells are removed from the paper and marked REMOVED_FROM_PAPER in the artifact graph.) All cells are re-read from the frozen `results_TITR*/TAIL_*` JSONs by `_bestrec_run/make_table_5_4_titration.py`, which **integrity-gates** the recomputed NDCG head/tail means against the locked prose values above (aborts on any $>5e-5$ drift); the HR columns and per-rung seed bands are the net-new tabulation the prose summarized only as Spearman trends.

ρ	kept inter./item	head $\Delta\text{NDCG}@10$	head $\Delta\text{HR}@10$	tail $\Delta\text{NDCG}@10$	tail $\Delta\text{HR}@10$
1.00	24.405	+0.002213 $\pm$ 0.000241 (5/5)	+0.004238 $\pm$ 0.000765 (5/5)	−0.000148 $\pm$ 0.000179 (2/5)	+0.000128 $\pm$ 0.000582 (3/5)
0.94	22.947	+0.002389 $\pm$ 0.000267 (5/5)	+0.004083 $\pm$ 0.000830 (5/5)	+0.000165 $\pm$ 0.000370 (4/5)	+0.000807 $\pm$ 0.000712 (5/5)
0.91	22.210	+0.002544 $\pm$ 0.000491 (5/5)	+0.004940 $\pm$ 0.000892 (5/5)	+0.000039 $\pm$ 0.000310 (3/5)	+0.000110 $\pm$ 0.000766 (3/5)
0.88	21.484	+0.002520 $\pm$ 0.000421 (5/5)	+0.004869 $\pm$ 0.000765 (5/5)	+0.000537 $\pm$ 0.000386 (4/5)	+0.000661 $\pm$ 0.000995 (4/5)
0.78	19.030	+0.002664 $\pm$ 0.000398 (5/5)	+0.004467 $\pm$ 0.001082 (5/5)	+0.000056 $\pm$ 0.000552 (2/5)	+0.000239 $\pm$ 0.001087 (2/5)
0.66	16.109	+0.003540 $\pm$ 0.000416 (5/5)	+0.005720 $\pm$ 0.000904 (5/5)	−0.000108 $\pm$ 0.000503 (1/5)	+0.000073 $\pm$ 0.001165 (3/5)

S.6 — Software V3 paired-seed detail

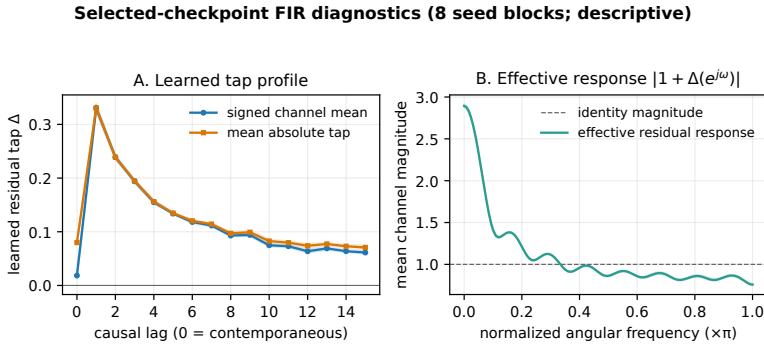
Software V3: outcome-known same-team robustness (8 matched seeds)



All points are registered sealed endpoints. Evidence class: exploratory; not independent confirmation.

Fig. S1. Software V3 registered paired-seed detail. Every matched seed block has higher sealed TEST NDCG@10 under learned FIR than identity; the second panel shows all paired differences, the frozen +0.000500 threshold, and the ordinary paired-t mean and 95% CI. This is outcome-known same-team robustness, not independent confirmation. Exact values are in `figures/fig_software_v3_pairs_data.csv`.

S.7 — Selected-checkpoint FIR diagnostics



Channels are summarized within seed; bands are ordinary 95% t intervals over eight seed summaries. Outcome-known control campaign; no mechanistic inference.

Fig. S2. Learned residual-tap and effective frequency-response summaries from the eight released selected checkpoints of the outcome-known active-control learned arm. Channels are summarized within each seed; bands are ordinary 95% t intervals over the eight seed summaries. Lag 0 is contemporaneous. These are descriptive fitted-operator diagnostics, not a mechanism or independent-confirmation test; plotted values are released in figures/fig_fir_response_data.csv. The effective kernel is the residual identity plus the learned Δ taps.

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